WERGONIC · MEASUREMENT & RISK CALCULATIONS

Possible Issues to Confirm

Twenty things found while mapping how the product turns sensor data into risk verdicts. **These are candidates, not confirmed defects.** Every one was read directly in the code and the facts stated below are accurate — but whether each is a *problem* depends on intent, and intent is not written down anywhere.

Each entry gives the facts, the case for it being a bug, the case for it being deliberate, and what to check to settle it. Nothing here should be sent to a developer as a ticket until the question in **"How to settle it"** has been answered.

20 items · 27 July 2026 · read against backend `staging`, mobile `main`, web `dev`
Two known-intentional deviations are excluded: the `-5°` lower edge on the "0°–10°" band, and trunk backward bending at `-10°` instead of RAMP's `0°`.

How to read the confidence labels

LIKELY A BUG The two sides contradict each other and no reading makes both correct. Still worth a sanity check, but expect this to become a ticket.

NEEDS A DECISION The behaviour is real and consistent, but whether it is right is a product or ergonomics judgement — not something the code can answer.

PROBABLY FINE Looks odd at first, and there is a plausible reason it is deliberate. Listed so it is not rediscovered later and mistaken for a bug.

Summary

ID	CONFIDENCE	ISSUE
P1	LIKELY A BUG	TNO analysis silently discards whole angle ranges
P2	LIKELY A BUG	A RAMP field, its comment and its label state three different thresholds
P3	LIKELY A BUG	Phone warns about head bending at 30° later than the reports do
P4	LIKELY A BUG	Phone and reports count backward bending differently
P5	LIKELY A BUG	Phone warns about arm speed at 60 °/s, reports warn at 30 °/s
P6	LIKELY A BUG	Mobile PDF and web screen print different recommended arm-speed limits
P7	LIKELY A BUG	Trunk and head bending percentiles never appear in the Word report
P8	LIKELY A BUG	RAMP has two colour functions that disagree at score 1.5
P9	LIKELY A BUG	MEC treats the same boundary value differently in different questions
P10	LIKELY A BUG	Phone instant feedback ignores backward bending entirely
P11	NEEDS A DECISION	Risk is a percentage of the recording, with no minimum length
P12	NEEDS A DECISION	Three different rules decide a colour for the same session

ID	CONFIDENCE	ISSUE
P13	NEEDS A DECISION	The web screen compares median speed against the yellow ceiling, not the green one
P14	NEEDS A DECISION	Trunk and head speed is measured and shown but never assessed
P15	NEEDS A DECISION	Head side bending is measured but has no risk band of its own
P16	NEEDS A DECISION	The printed criteria text states ranges that overlap
P17	NEEDS A DECISION	Temperature is printed with no unit in two of the three reports
P18	PROBABLY FINE	Pie charts group arm angles differently from the risk gauges
P19	PROBABLY FINE	The “Extension” slice and the “backward” risk row measure different populations
P20	PROBABLY FINE	P99 appears in some reports and not others

THE PATTERN UNDERNEATH

Six of these exist because the thresholds are written down three times.

The backend has a file that calls itself the single source of truth and says the numbers *“can never drift apart again”*. The web app then re-types those numbers by hand into its own file. The mobile app hardcodes its own copies inside each feedback class.

Two hand-made copies of a single source of truth is not a single source of truth. **P3, P4, P5, P6, P8 and P9 are all the same root cause** — a threshold was updated in one copy and not the others. Fixing them individually leaves the mechanism intact and they will drift again.

Likely a bug

The two sides contradict each other, and no reading of the code makes both correct.

P1 TNO analysis silently discards whole angle ranges LIKELY A BUG

What happens	The TNO calculation sorts each movement into an angle range. One range in the middle is never filled, so any movement landing there is dropped and counted nowhere.
For arms	Movements between **10° and 20°** are discarded.
For the trunk	Movements between **0° and 20°** are discarded — which is most normal working posture.
Why	The condition that selects the middle range is `0 < i < length - 1`. The trunk is configured with only two boundaries, so this becomes `0 < i < 1` — impossible for any whole number.
Where	`tno/calculations/extractTaskCharacteristics.py:40-42`

CASE FOR IT BEING A BUG

The range exists in the output structure, has a label, and is read by the mobile app — which expects to receive it. Nothing suggests the range was meant to be empty. It reports **0%** even when the worker spent the whole shift there, which is not a missing feature but a wrong number.

CASE FOR IT BEING DELIBERATE

Weak. Somebody could have decided the middle range is uninteresting — but then it would not be defined, labelled and consumed downstream. No comment anywhere says so.

HOW TO SETTLE IT Open any completed session and look at the TNO table. If the middle range reads exactly `0%` on every session ever recorded, it is the bug. This takes one minute and is the strongest item on the list.

P2 A RAMP field, its comment and its label state three different thresholds LIKELY A BUG

The code does	Counts time where head bending is **−5° or further back**.
The comment says	“for values ≤ **−10**”.
The field is named	`backward_bending_gt_10_degrees`.
The customer reads	“Percentage of time with backward bending **more than 10 degrees**”.
Where	`ramp/calculations/ramp_calculator.py:273-282`

CASE FOR IT BEING A BUG

Three statements about one number, in ten lines, none of which agree. At minimum the comment and the customer label are wrong. If the intended threshold really is −10°, the calculation is wrong too and every RAMP head score is slightly overstated.

CASE FOR IT BEING DELIBERATE

The *value* −5° may well be correct and intentional — it matches the −5° edge used elsewhere, which is confirmed as deliberate. In that case this is purely a naming and labelling problem, not a calculation one.

HOW TO SETTLE IT One question to whoever specified RAMP: **should head backward bending be measured from −5° or from −10°?** Note the risk gauges elsewhere in the product use −10° for the same body part, which is an argument that −10° was intended here too.

P3 Phone warns about head bending later than the reports do LIKELY A BUG

The phone warns at	**20°** (orange) and **45°** (red)
The reports warn at	**30°**
What the worker sees	A worker holding their head at 35° forward sees **no warning** during the recording, then the report marks it as a risk factor afterwards.
Why it looks accidental	The head feedback class is a line-for-line copy of the trunk one. **20° and 45° are the trunk's numbers.** The head's tighter limits were never substituted in.
Where	Phone: `head_feedback_manager.dart:18-23` · Reports: `posture_thresholds.py:19`

CASE FOR IT BEING A BUG

The neck has far less range of movement than the lower back, which is exactly why the reports use 30° for the head and 45° for the trunk. Applying the trunk's numbers to the neck contradicts our own risk model, and the copy-paste origin is visible in the code.

CASE FOR IT BEING DELIBERATE

Live feedback could be intentionally less sensitive than the report, so workers are not buzzed constantly. But if so it should be a deliberate offset, not the trunk's numbers by accident, and it should apply to every body part rather than just this one.

HOW TO SETTLE IT Ask whether live warnings are meant to fire at the same angle as the report. If yes, this is a bug. If no, we need a stated rule for how much less sensitive live feedback should be.

P4 Phone and reports count backward bending differently LIKELY A BUG

The phone counts	**any** backward lean at all (anything below 0°), then warns when that exceeds **6%** of the session and alarms above **12.5%**.
The reports count	only leans past **−10°**, then warn above **1%** and alarm above **6%**.
The mismatch	Both the angle and the percentage differ. The phone counts far more moments as "backward", but then requires a much larger share before reacting.
Where	Phone: `bending_feedback_manager.dart:22-23, 35-41` · Reports: `posture_thresholds.py:18,20`

CASE FOR IT BEING A BUG

The phone is applying one rule's counting method to another rule's limits. There is no version of the risk model in which "any lean past 0°, warn above 6%" is a rule we chose — the number 0 appears nowhere in our threshold files, and the 6% figure is the report's *alarm* level being used as the phone's *warning* level.

CASE FOR IT BEING DELIBERATE

Possible that whoever wrote it reasoned the two errors cancel out — counting more moments but requiring a higher share. That is not a safe assumption and nothing documents it.

HOW TO SETTLE IT Worth measuring rather than reasoning about: take one real session and compute the backward-bending percentage both ways. If they land in different colours, this is a bug regardless of intent.

P5 Phone warns about arm speed at a different threshold from the reports LIKELY A BUG

The phone warns at	**60 °/s**
The reports warn at	**15 °/s** (yellow) and **30 °/s** (red)
The decisive fact	Our own Word report explains, in text shown to customers, that the original standard used **60 °/s** and we deliberately changed it to **30 °/s** because IMU sensors estimate more accurately than the accelerometer-only sensors the original research used.
Where	Phone: `arm_feedback_manager.dart:84-91` · Reports: `posture_thresholds.py:25` · Rationale: `session_docx_report.py:1171-1176`

CASE FOR IT BEING A BUG

60 is the number we publicly say we no longer use. The backend was updated when the decision was made; the phone was not. This is the clearest example of the copy-drift problem on the list.

CASE FOR IT BEING DELIBERATE

One genuine complication: the phone and the backend measure arm speed by different methods. The phone uses total rotation in every direction; the backend uses only the rate at which the arm rises and falls. The phone's number is therefore always the larger of the two, which could justify a higher threshold — but nobody has written down that this is why, and 60 matching the superseded standard exactly is hard to call a coincidence.

HOW TO SETTLE IT Ask whoever made the 60 → 30 change whether the phone was in scope. Also worth deciding whether the phone and the reports should be measuring arm speed the same way at all — see P12.

P6 The mobile PDF and the web screen print different recommended limits LIKELY A BUG

Mobile PDF prints	Recommended limit for median arm speed: **20 °/s**
Web screen prints	Recommended limit for median arm speed: **30 °/s**
Canonical file says	Green up to **15**, yellow up to **30**
Where	Mobile: `pdf_table_report.widget.dart:555` (hardcoded text) · Web: `MovementSectionV2.tsx:362`

CASE FOR IT BEING A BUG

Three numbers for one limit, and the customer is told two of them as advice. **20 appears in no threshold file anywhere** — it is not the green cutoff, not the yellow cutoff, and not the old standard. It looks like a typed literal that was never traced to anything.

CASE FOR IT BEING DELIBERATE

None found. Even if 20 were a considered value, it is not stored anywhere it could be maintained from.

HOW TO SETTLE IT Decide which number is the recommended limit for the median — 15 or 30 — then make both reports read it from the canonical file instead of printing a literal. See also P13, which questions whether 30 is the right choice.

P7 Trunk and head bending percentiles never appear in the Word report LIKELY A BUG

What is computed	The backend calculates and stores the 10th, 50th, 90th and 99th percentiles for trunk and head forward bending, side bending, and arm elevation.
What is printed	The Session Percentiles table only knows how to print two rows per body part: *Elevation Angle* and *Angular Velocity*.
The consequence	Trunk and head have no elevation angle — they have forward bending. So their bending percentiles are silently omitted, and any trunk or head row printed under "Elevation Angle" would be empty.
Where	`session_docx_report.py:530-584`

CASE FOR IT BEING A BUG

We do the work of calculating these numbers and then throw them away at the last step. The Word report is our most complete artefact and the one we hand to customers, so this is where the omission costs most.

CASE FOR IT BEING DELIBERATE

Possible the section was scoped to arms on purpose and never widened. But the table iterates over every body part, which suggests the intent was to cover them all.

HOW TO SETTLE IT Open a recent Word report and look at the Session Percentiles table. If there are rows labelled "Trunk Elevation Angle" filled with dashes, that confirms it.

P8 RAMP has two colour functions that disagree LIKELY A BUG

Function one says	Score **1.5** → **yellow**
Function two says	Score **1.5** → **green**
Both are live	They run on different screens of the RAMP results, for the same posture fields.
Is 1.5 reachable?	Yes. It is a real output of the head-backward score ladder.
Where	`RAMP/utils/functions.ts:321-341` and `RAMP/utils/generatedCalculations.ts:9-33`

CASE FOR IT BEING A BUG

The same score renders two different colours in two places in one assessment. Only one can be right.

CASE FOR IT BEING DELIBERATE

One function may have been written for the auto-generated view and the other for the manually completed view, with genuinely different colour rules intended. If so, that should be stated, because it is not visible from either screen.

HOW TO SETTLE IT Decide the correct colour for score 1.5, then delete one of the two functions. Note: an earlier version of this finding claimed scores between 2 and 3 render green — **that was wrong**. RAMP scores are discrete and nothing lands between 2 and 3. The disagreement at 1.5 is the real one.

P9 MEC treats the same boundary value differently in different questions LIKELY A BUG

Question 1 says	Exactly **8 minutes per hour** of static time → **red**
Questions 9, 10, 11 say	Exactly **8 minutes per hour** of static time → **yellow**
Same test	Both use the same 4 and 8 minute cutoffs on the same measurement. Only the comparison differs — one includes the boundary, the other excludes it.
Where	`MEC/utils/calculations.ts:120-126` versus `:214-218`

CASE FOR IT BEING A BUG

There is no reason one body part should treat 8.00 differently from another. No comment explains it, and the frequency thresholds in the same function already use the other convention consistently.

CASE FOR IT BEING DELIBERATE

None found — but the practical impact is close to zero. These are decimal minutes; landing on exactly 8.00 is vanishingly rare with real sensor data.

HOW TO SETTLE IT Not urgent, but do not leave it deliberately. It is a one-character fix. **The cost is credibility, not accuracy** — MEC is the assessment we can call fully objective, and two questions disagreeing on identical inputs undermines that claim if anyone notices.

P10 Phone instant feedback ignores backward bending entirely LIKELY A BUG

Instant feedback mode	Only reacts to forward bending. Any backward lean, however extreme, shows **grey**.
Accumulated feedback mode	Does count backward bending and will turn orange or red because of it.
The reports	Treat backward bending as a named risk factor with its own thresholds.
Where	`bending_feedback_manager.dart:48-59`

CASE FOR IT BEING A BUG

The same app gives contradictory guidance depending on which feedback mode the worker has selected. Backward bending is a real risk factor everywhere else in the product.

CASE FOR IT BEING DELIBERATE

Instant feedback is meant to be simple and immediate; adding a backward rule may have been judged confusing. Plausible — but then the accumulated mode should not contradict it.

HOW TO SETTLE IT Decide whether instant feedback should cover backward bending. Whichever way, the two modes should agree.

Needs a decision

The behaviour is real and consistent. Whether it is *right* is a product or ergonomics judgement.

P11 Risk is a percentage of the recording, with no minimum length NEEDS A DECISION

How it works	Every risk colour is "what share of this recording was spent in the bad posture". There is no adjustment for how long the recording was, and no minimum length.
Consequence one	A 10-minute recording with 5 minutes of overhead work scores **50% — red**. A 10-hour recording with 4 hours of overhead work scores **40% — yellow**. Four hours rates better than five minutes.
Consequence two	A 2-minute recording where the worker reaches up to a shelf for 20 seconds scores **17% above 60° — red**. For one shelf.
Verified	No normalisation to a standard workday and no minimum-duration guard exists anywhere in the backend.

CASE FOR IT BEING A PROBLEM

As implemented, the same physical exposure produces opposite verdicts depending on how long we happened to record. A customer who notices this will not trust the tool.

CASE FOR IT BEING CORRECT

Genuine and important. The research we cite gives limits *for an eight-hour workday*, and the method assumes a recording is a **representative sample** of the shift. Under that assumption percentage is exactly the right unit and duration is irrelevant. The maths is not wrong — the assumption is simply never enforced or stated.

HOW TO SETTLE IT Three cheap options, not mutually exclusive: (1) set a minimum recording length below which the colour is suppressed or marked indicative; (2) show the absolute time next to the percentage — **we already calculate it and it is already in the tables**, the colour just ignores it; (3) print the assumption in the report: "assumes this period is representative of the full shift."

P12 Three different rules decide a colour for the same session NEEDS A DECISION

Rule one	**Share of time in a bad posture.** Used by the Word report and the web risk gauges. This is the main model.
Rule two	**A percentile compared against a fixed limit**, with a warning band at 80% of the limit. Used only on the web measurement screen. No other report applies it.
Rule three	**The angle itself.** The Appendix pie charts colour arms green, yellow and red at 20° and 60°.
Consequence	A customer comparing the pie chart, the risk gauge and the on-screen view sees three colour schemes for the same arm in the same session.

CASE FOR IT BEING A PROBLEM

None of the three is wrong on its own, but together they make the product look like it cannot decide what "risky" means. The 80% warning band in particular exists in exactly one place and is documented nowhere.

CASE FOR IT BEING CORRECT

They answer different questions and all three are legitimate: how long, how high, and what the spread looks like. A single number cannot express all three.

HOW TO SETTLE IT This is presentation, not maths. Decide which rule is *the* verdict and present the others as supporting detail with different visual treatment, so three colour schemes stop competing for the same meaning.

P13 The web screen compares median speed against the yellow ceiling, not the green one NEEDS A DECISION

Canonical bands	Arm speed: green up to **15 °/s**, yellow up to **30 °/s**, red above.
What the screen does	Shows the recommended limit for median arm speed as **30 °/s** — the ceiling of the yellow band.
Effect	A worker whose median arm speed is 25 °/s is already in the yellow band by the canonical model, but the screen shows them comfortably under the stated limit.

CASE FOR IT BEING A PROBLEM

Presenting the top of the caution band as "the limit" makes a moderate-risk worker look compliant.

CASE FOR IT BEING CORRECT

Defensible if the intent is "the limit you must not exceed" rather than "the level you should stay under". The 80% warning band partly compensates, flagging yellow from 24 °/s.

HOW TO SETTLE IT Decide what the word "limit" means in this table — the green ceiling or the red floor — and make it consistent with P6.

P14 Trunk and head speed is measured and shown but never assessed NEEDS A DECISION

What exists	Trunk and head rotation speed is captured by the sensors, stored, and printed in report columns.
What is missing	No threshold exists for it. Only arm and wrist speed have bands, so these columns can never be coloured.

CASE FOR IT BEING A PROBLEM

A column of numbers a customer cannot act on. Either it means something or it should not be shown.

CASE FOR IT BEING CORRECT

Likely deliberate: the research we cite gives velocity limits for arms and wrists, not for the trunk or neck. We may simply have no defensible threshold to apply.

HOW TO SETTLE IT Ask whether a published trunk velocity limit exists. If not, either drop the columns or label them explicitly as informational.

P15 Head side bending is measured but has no risk band of its own NEEDS A DECISION

What exists	Head side bending is measured and shown as a distribution, in the same 0–10° / 10–30° / above 30° ranges as the trunk.
What is missing	The risk table covers arm elevation, trunk forward, trunk backward, head forward and head backward — and stops. Head side bending can never turn red.
Where it does count	RAMP uses head side bending past **10°** as a trigger for its first posture question. So it affects a RAMP score but never a session risk gauge.

CASE FOR IT BEING A PROBLEM

Inconsistent: it is good enough to drive a RAMP score but not good enough to appear in the risk summary. If sideways neck strain matters in one place it should matter in the other.

CASE FOR IT BEING CORRECT

The same is true of trunk side bending, which suggests a deliberate scope: the risk gauges cover the factors the cited research gives eight-hour limits for, and side bending may not be among them.

HOW TO SETTLE IT Ask whether the omission of side bending from the risk gauges was a decision or an oversight. It applies to both head and trunk, so it is one question, not two.

P16 The printed criteria text states ranges that overlap NEEDS A DECISION

The report prints	`0 – 5 % = Low risk` · `5 – 10 % = Moderate risk` · `10 – 100 % = High risk`
The code means	Up to and including 5% is low · above 5% and up to 10% is moderate · above 10% is high
The problem	As printed, 5% appears in two bands and 10% appears in two bands. A reader cannot tell which one applies.

CASE FOR IT BEING A PROBLEM

This is the text a customer's safety officer reads to understand our verdicts. Ambiguous boundaries in a compliance-adjacent document invite exactly the wrong kind of question.

CASE FOR IT BEING CORRECT

The calculation is right; only the wording is loose. In practice a value landing exactly on a boundary is rare.

HOW TO SETTLE IT Purely editorial. Print `≤ 5 %`, `> 5 – 10 %`, `> 10 %`. It is generated from one function, so it is a single change affecting every criteria block.

P17 Temperature is printed with no unit in two of the three reports NEEDS A DECISION

Word report	Prints bare numbers, e.g. `32.45`
Web PDF	Also prints bare numbers
Web screen	Correctly prints `32.45 °C`

CASE FOR IT BEING A PROBLEM
An unlabelled temperature is ambiguous to any reader outside a Celsius country, and the on-screen view already proves we know the unit.

CASE FOR IT BEING CORRECT
Context makes it obvious to most readers, and no customer has raised it.

HOW TO SETTLE IT Trivial fix, no decision really needed beyond confirming the unit is Celsius. Bundle with P16.

Probably fine

Looks odd, has a plausible reason. Recorded so it is not rediscovered later and mistaken for a defect.

P18 Pie charts group arm angles differently from the risk gauges PROBABLY FINE

Pie charts group at	**20°** and **60°**
Risk gauges use	time above **30°** and time above **60°**
Why it is probably fine	The pie chart is a picture of how the session was distributed; the gauge is a verdict. They answer different questions and neither claims to be the other.
Why it still matters	The pie slices are coloured green, yellow and red — the same colours as the verdict. That is what invites the comparison.

HOW TO SETTLE IT If left as is, consider using neutral colours for the distribution chart so it stops reading as a second opinion on the risk level.

P19 The "Extension" slice and the "backward" risk row measure different populations PROBABLY FINE

The Extension slice counts	anything past **−5°**
The backward risk row counts	anything past **−10°**
Why it is probably fine	Both numbers are confirmed intentional. The −5° edge exists to absorb sensor noise in the distribution view; the −10° threshold is the documented, deliberate modification of the RAMP standard for the risk verdict. The risk file even carries a comment saying to use −10° consistently.
Why it still matters	The two appear in the same document and sound like the same thing, so they will show different percentages for what a reader assumes is one measurement.

HOW TO SETTLE IT No code change needed. Consider labelling the slice "Extension (past −5°)" and the risk row "Backward (past −10°)" so the difference is visible rather than surprising.

`P20` P99 appears in some reports and not others PROBABLY FINE

Shows P99	Word report, web measurement screen
Does not	Mobile PDF, web PDF (which has no percentiles table at all)
Why it is probably fine	Shorter reports omitting the most technical column is a reasonable editorial choice, and P99 is the least actionable of the four.

HOW TO SETTLE IT Only worth changing if customers are comparing reports and asking why a column is missing. Otherwise leave it.

If you only do three things

Ranked by what it costs us to leave alone

1. Confirm P1 (TNO). One minute of looking at a report settles it. If confirmed, we are reporting 0% for a range where workers spend most of their time, and every TNO figure we have ever shown a customer is affected.

2. Decide P11 (percentage without duration). Not a bug, but the one item on this list a customer could independently work out and challenge us on. Showing absolute time next to the percentage costs almost nothing because we already calculate it.

3. Fix P3, P4, P5 and P6 as one job. They are four symptoms of one cause — the phone holds its own hardcoded copies of thresholds that have since moved. Fix them individually and they will drift again; give the phone one shared source and they cannot.

Read against backend `staging` @ `7153485`, mobile `main` @ `3726867e`, web `dev` @ `83a43d4f`. Every fact stated was read directly in the source. The confidence labels are judgements about intent, not about the facts.